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Chassis component having a receptacle for an elastomer bearing

Abstract

A chassis component having a receptacle for an elastomer bearing(s), the receptacle has a first cylinder section and a second cylinder section. The first cylinder section is formed by a rim hole and the second cylinder section is formed by a sleeve. The chassis component is formed from a first and a second sheet metal shell, and the sleeve is joined materially to the second sheet metal shell.

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Background/Summary

RELATED APPLICATIONS

(1) The present application claims priority of German Application Number 10 2022 107 298.3 filed Mar. 28, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

FIELD

(2) The disclosure relates to a chassis component.

(3) The chassis is part of a motor vehicle. The chassis is responsible for passenger comfort and makes a contribution to driving dynamics and safety by ensuring that the forces generated in the contact zones between the road and the tire are transmitted in an optimum manner. The chassis is a complex combination of chassis components which serve to connect the body to the roadway.

BACKGROUND

(4) In motor vehicles, chassis components are connected to the body, drive components or other chassis components via bearing(s). Elastomer bearing(s) are used in chassis assemblies for motor vehicles. They are mounted at the connection points to the vehicle body, to the transmission, to the drive unit or to chassis links in order, for example, to decouple the vibration or noises of the differential. Elastomer bearing(s) are able to absorb forces by deformation of the elastomer and are pressed by means of the outer sleeve into a correspondingly prepared receptacle of the chassis component.

(5) Sleeves are used to receive the elastomer bearing(s), as described by DE 10 2019 131 716 B4. The sleeves are able to be made, for example, from a tube section or are able to be designed as a slotted or rolled sleeve. The sleeves are joined to the chassis component or are riveted or screwed to the chassis component in the form of separate parts, for example, as a forged part. In order to join the sleeves materially, a sufficient free space, e.g., for a welding unit, should be ensured and a corresponding projection for the formation of the welded seam should additionally be provided

structurally. In the case of complex installation spaces, said configuration leads to design limitations.

(6) Alternatively, the receptacle is able to be designed as a double-sided rim hole, as shown in DE 10 2016 107 155 A1 and DE 10 2011 052 398 B4. A rim hole in a workpiece is able to be formed, for example, by a punching, drilling, roller forming or spreading method. In the case of a rim hole, connection by means of joining is therefore eliminated. The execution of the rim hole is demanding in terms of production technology and limited to a maximum depth of the rim hole collars. This is a disadvantage compared with the sleeve solution, in which the depth of the receptacle is not limited. In the chassis sector, rim hole depths of between 18 mm and 20 mm are nowadays common.

Elastomer bearing(s) are able to have a height of more than 60 mm. The rim hole thus limits the height of the elastomer bearing(s), or a distance is created between the rim holes which is able to adversely affect the secure retention of the elastomer bearing(s) and lead to manufacturing problems. Owing to the fact that the rim hole collars cannot be deformed to an arbitrary depth, the friction area with respect to the elastomer bearing(s) and thus the maximum effective press-out force is reduced.

(7) DE 10 2015 200 352 A1 describes a link for a wheel suspension of a vehicle having a receptacle for an elastomer body. This publication is able to be referred to as defining the type in question.

(8) DE 10 2015 210 915 A1 also describes a link for a wheel suspension of a vehicle having a bearing(s) receptacle. The receptacle is formed by a bushing and two opposing rim holes.

(9) DE 10 2014 222 577 A1 describes a control arm with a ball joint in press-fit connection.

SUMMARY

(10) At least one embodiment of the disclosure provides a chassis component having an improved receptacle for an elastomer bearing(s), which ensures a large friction area between the elastomer bearing(s) and the receptacle and doing so with as few joining operations as possible, thus enabling the receptacle to be implemented in installation spaces with limited free space.

(11) According to at least one embodiment of the disclosure, a chassis component includes a receptacle for an elastomer bearing(s).

(12) A chassis component is, for example, a subframe, longitudinal member, crossmember or A arm. The receptacle of the chassis component has a first and a second cylinder section, wherein the first cylinder section is formed by a rim hole and the second cylinder section is formed by a sleeve.

(13) The chassis component according to at least one embodiment of the disclosure is formed from a first and a second sheet metal shell. The two sheet metal shells are assembled to form the chassis component and joined to or along the contacting limbs. The sheet metal shells form a hollow profile, at least in some location or locations. The rim hole is a materially unitary part of a first sheet metal shell. According to at least one embodiment of the disclosure, the sleeve is joined materially to a second sheet metal shell. The sleeve is able to be designed, for example, as a slotted, perforated, multi-layered rolled or longitudinally joined sheet/tube.

(14) The rim hole and the sleeve are arranged on a common central longitudinal axis. The end faces of the inner end of the sleeve and of the collar of the rim hole are situated opposite one another so as to rest against one another. The clearance is between 0.1 and 10 mm, between 0.2 and 5 mm and between 0.3 and 3 mm.

(15) The structural length of the receptacle is able to be adapted to the elastomer bearing(s), such as by adapting the sleeve length or the depth of the rim hole. The receptacle is able to be adapted to the height of the elastomer bearing(s) by means of the distance between the end faces of the sleeve and the collar.

(16) The combination of a sleeve and a rim hole ensures a large friction area of the receptacle with the elastomer bearing(s). At the same time, locations which are critical in terms of installation space since are able to be responded to, in comparison with the pure sleeve solution, one of the two joining operations is able to be omitted. The sleeve becomes shorter and thus less expensive than the pure sleeve solution. The rim hole is able to be executed at a depth that is non-critical in terms

of production technology and cost-effective. The combination of a sleeve and a rim hole is able to be used for many receptacles of chassis components, such as for subframes and A arms and control arms.

(17) In at least one embodiment of the receptacle, the sleeve is joined, for example welded, to the second sheet metal shell over an entire circumference of the sleeve in the region of an opening of the second sheet metal shell. The opening is circular or oval. Alternatively, the sleeve also is able to be joined materially to the second sheet metal shell in the region of a part-circular opening of the second sheet metal shell, for example a semicircle, such as by means of a weld seam and at the end of a control arm leg. The opening of the second sheet metal shell is able to be arranged in a recess, thus ensuring that the outer end of the joined sleeve does not project beyond the surface of the second sheet metal shell, or terminates flush with the second sheet metal shell. In this context, the surface of the second sheet metal shell is understood to be the section around the elastomer bearing(s). This notwithstanding, the second sheet metal shell also is able to be designed to be recessed in a section at least 200 mm away from the elastomer bearing(s), for example, at an opposite end of the chassis component.

(18) In at least one embodiment of the receptacle, the inner end of the sleeve and the collar of the rim hole are able to overlap in a certain region and, for example, are able to be connected to one another in a non-positive and/or positive manner. The receptacle is able to be adapted to the height of the elastomer bearing(s) by means of the height of the overlap between the sleeve and the collar.

(19) Sheets of a steel alloy are used as the material for the chassis component. The sleeve is made of a material different from the sheet metal shells, but also of a steel material. The sheet metal shells also is able to have tailor-made properties, for example, in respect of mechanical characteristics or wall thicknesses. The chassis component is painted. The chassis component is able to have a corrosion protection coating.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure is described below with reference to drawings. In the drawings:
- (2) FIG. 1 shows a chassis component as part of a subframe in a perspective view according to at least one embodiment of the disclosure;
- (3) FIG. 2 shows a perspective front view of a receptacle according to at least one embodiment of the disclosure;
- (4) FIG. 3 shows a front view of the receptacle according to at least one embodiment of the disclosure;
- (5) FIG. 4 shows a front view of the receptacle according to at least one embodiment of the disclosure;
- (6) FIG. 5 shows a front view of the receptacle according to at least one embodiment of the disclosure; and
- (7) FIG. 6 shows a perspective view of the receptacle according to at least one embodiment of the disclosure.

DETAILED DESCRIPTION

- (8) FIG. 1 shows chassis components in the form of two longitudinal members 1, 2, which are part of a subframe 3. The longitudinal members 1, 2 are connected to one another by two crossmembers 4, 5.
- (9) The chassis components are formed from a main body 6, 7. The main body 6, 7 has two sheet metal shells 8, 9. The sheet metal shells 8, 9 are formed by pressing and are arranged one above the other in the image plane with the exception of overlapping joining regions.
- (10) A receptacle 11 according to at least one embodiment of the disclosure is arranged in each case

at a rear end **10** of the main body **6, 7** in the direction of travel of the motor vehicle. A bearing(s) receptacle **13** is provided at a front end **12**. In FIG. 1, the bearing(s) receptacles **13** are embodied as long sleeves **14**. The long sleeves **14** are joined to the sheet metal shells **8, 9** of the main body **6, 7**. According to at least one embodiment of the disclosure, the receptacles **11** have a first cylinder section **15** and a second cylinder section **16**, wherein the first cylinder section **15** is formed by a rim hole **17** in the first sheet metal shell **8** and the second cylinder section **16** is formed by a sleeve **18**. The sleeve **18** of the second cylinder section **16** is joined to the second sheet metal shell **9**.

(11) FIG. 2 shows the detail view of a receptacle **11** according to at least one embodiment of the disclosure for an elastomer bearing(s). The rim hole **17** and the sleeve **18** are arranged on a common central longitudinal axis **19**. The inner end **20** of the sleeve **18** and the collar **21** of the rim hole **17** rest on the end face. The height **h1** of the receptacle **11** is able to be adapted to the elastomer bearing(s) by means of the height **h2** of the collar **21** of the rim hole **17** and the height **h3** of the sleeve **18**.

(12) FIG. 3, FIG. 4, FIG. 5, and FIG. 6 show some embodiments of the receptacle **22, 23, 24, 25** for the elastomer bearing(s).

(13) FIG. 3 shows at least one embodiment of the disclosure in the form of the receptacle **22**, in which the inner end **20** of the sleeve **18** and the collar **21** of the rim hole **17** overlap in a certain region and are connected to one another in a non-positive manner. The height **h4** of the receptacle **22** is able to be adapted to the elastomer bearing(s) by means of the height **h5** of the overlap.

(14) FIG. 4 shows at least one embodiment of the disclosure in the form of the receptacle **23**, in which the end faces of the inner end **20** of the sleeve **18** and of the collar **21** of the rim hole **17** are situated opposite one another. The height **h6** of the receptacle **23** is able to be adapted to the elastomer bearing(s) by way of the clearance **h7** between the inner end **20** of the sleeve **18** and the collar **21** of the rim hole **17**.

(15) As shown in FIG. 5, at least one embodiment of the disclosure the opening **26** of the second sheet metal shell **9** is able to be arranged in a recess **27**, thus ensuring that the outer end **28** of the joined sleeve **18** does not project beyond the surface **29** of the second sheet metal shell **9**.

(16) FIG. 6 shows at least one embodiment of the disclosure, in which the sleeve **18** is joined materially to the second sheet metal shell **9** by means of a semicircular weld seam in the region of a semicircular opening **30** of the second sheet metal shell **9**.

(17) The foregoing description of some embodiments of the disclosure has been presented for purposes of illustration and description. The description is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings. The specifically described embodiments explain the principles and practical applications to enable one ordinarily skilled in the art to utilize various embodiments and with various modifications as are suited to the particular use contemplated. Various changes, substitutions and alterations can be made hereto without departing from the spirit and scope of the disclosure.

Claims

1. A chassis component, comprising: a receptacle for an elastomer bearing; a first sheet metal shell connected to the receptacle; and a second sheet metal shell connected to the receptacle, the receptacle comprising a first cylinder section and a second cylinder section, the first cylinder section comprising a rim hole and the second cylinder section comprising a sleeve, wherein the sleeve is welded to the second sheet metal shell, wherein the rim hole projects inwardly from the first sheet metal shell in a direction towards the sleeve and the second sheet metal shell, an end face of the sleeve and an end face of a collar of the rim hole are situated opposite one another, the rim hole and the sleeve are on a common central longitudinal axis, and along the common central longitudinal axis, an entirety of the sleeve is spaced from an entirety of the collar of the rim hole.

2. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell over a circumference in entirety of the sleeve at a region of an opening of the second sheet metal shell.
 3. The chassis component according to claim 1, wherein the rim hole is a materially unitary part of the first sheet metal shell.
 4. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell in a region of an opening of the second sheet metal shell.
 5. The chassis component according to claim 4, wherein the opening of the second sheet metal shell is in a recess of the second sheet metal shell.
 6. The chassis component according to claim 4, wherein an outer end of the sleeve does not project beyond a surface of the second sheet metal shell.
 7. The chassis component according to claim 1, wherein the sleeve is welded to the second sheet metal shell in a region of a part-circular opening of the second sheet metal shell.
 8. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole overlap one another when viewed from an overhead perspective of the chassis component.
 9. The chassis component according to claim 1, wherein the chassis component is a link.
 10. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole are connected to one another in a non-positive manner.
 11. The chassis component according to claim 1, wherein an inner end of the sleeve and the collar of the rim hole are connected to one another in a positive manner.
 12. The chassis component according to claim 1, wherein the chassis component is a subframe of a motor vehicle or a part thereof.
 13. The chassis component according to claim 1, wherein the chassis component is an engine frame of a motor vehicle or a part thereof.
 14. The chassis component according to claim 1, wherein along the common central longitudinal axis, the end face of the sleeve faces toward, and is spaced from, the end face of the collar of the rim hole.
 15. The chassis component according to claim 14, wherein the entirety of the sleeve is outside the rim hole.
 16. A chassis component, comprising: a receptacle for an elastomer bearing; a first sheet metal shell connected to the receptacle; and a second sheet metal shell connected to the receptacle, the receptacle comprising a first cylinder section and a second cylinder section, the first cylinder section comprising a rim hole and the second cylinder section comprising a sleeve, wherein the sleeve is welded to the second sheet metal shell, an end face of the sleeve and an end face of a collar of the rim hole are situated opposite one another, the rim hole and the sleeve are on a common central longitudinal axis, and along the common central longitudinal axis, an entirety of the sleeve is spaced from an entirety of the collar of the rim hole.
 17. The chassis component according to claim 16, wherein the sleeve is welded to the second sheet metal shell over a circumference in entirety of the sleeve at a region of an opening of the second sheet metal shell.
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